



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2

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SECP2613-02 SYSTEM ANALYSIS AND DESIGN (WBL)

SECTION 02

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CONTEXT DIAGRAM (PROPOSED SYSTEM)

Figure 1 illustrates the proposed solution for managing loan applications and memberships within the Lembaga Koperasi Kakitangan KADA System. The system acts as a central hub, dealing with the main entities: applicants, administrators, and Ahli Lembaga Koperasi. Applicants submit registration forms, loan applications, membership payments, and personal information to the system, which in turn issues a user ID. The Admin receives loan information and applicant forms from the system, reviews the data, and sends back reviewed information and loan statements. Ahli Lembaga Koperasi receives applicant information and membership approval details from the system, decides on loan approvals and communicates the approval status back to the system.

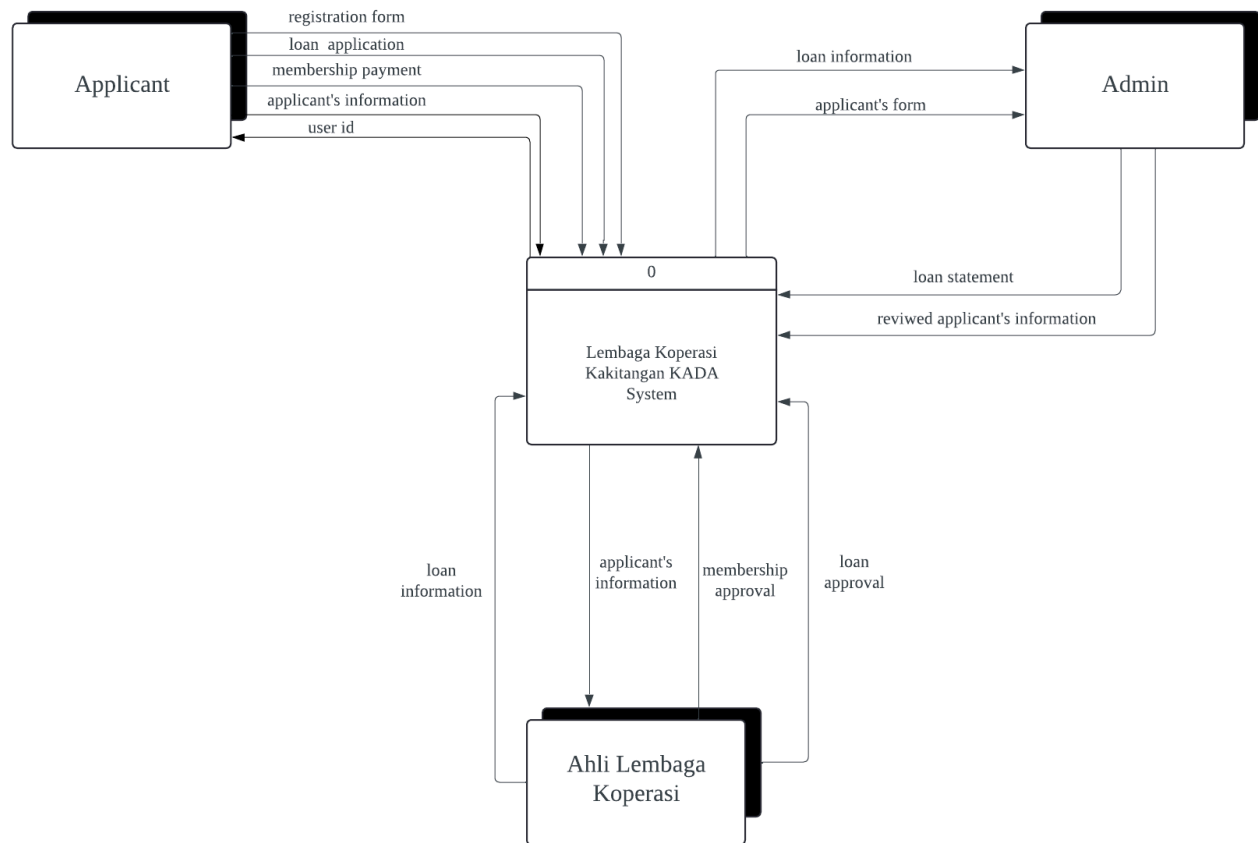


Figure 1: Context Diagram

LEVEL 0 DIAGRAM (PROPOSED SYSTEM)

Figure 2 for the proposed membership and loan application system displays the main processes and their relationships. Applicants log into the system to apply for membership and provide their details, which are recorded in the Applicant Data database (D1). The admin reviews these forms to validate the information. Once validated, the membership is approved and the status is notified to the applicant. For loan applications, the applicant completes an online form, which is then used to calculate the loan and dividend data. The loan applications are reviewed by the administrator and Ahli Lembaga Koperasi, and once approved, the information is recorded in the Loan Database (D2). The final loan calculation is communicated back to the applicant. Additionally, the system generates and sends notifications via email for updates on membership and loan status.

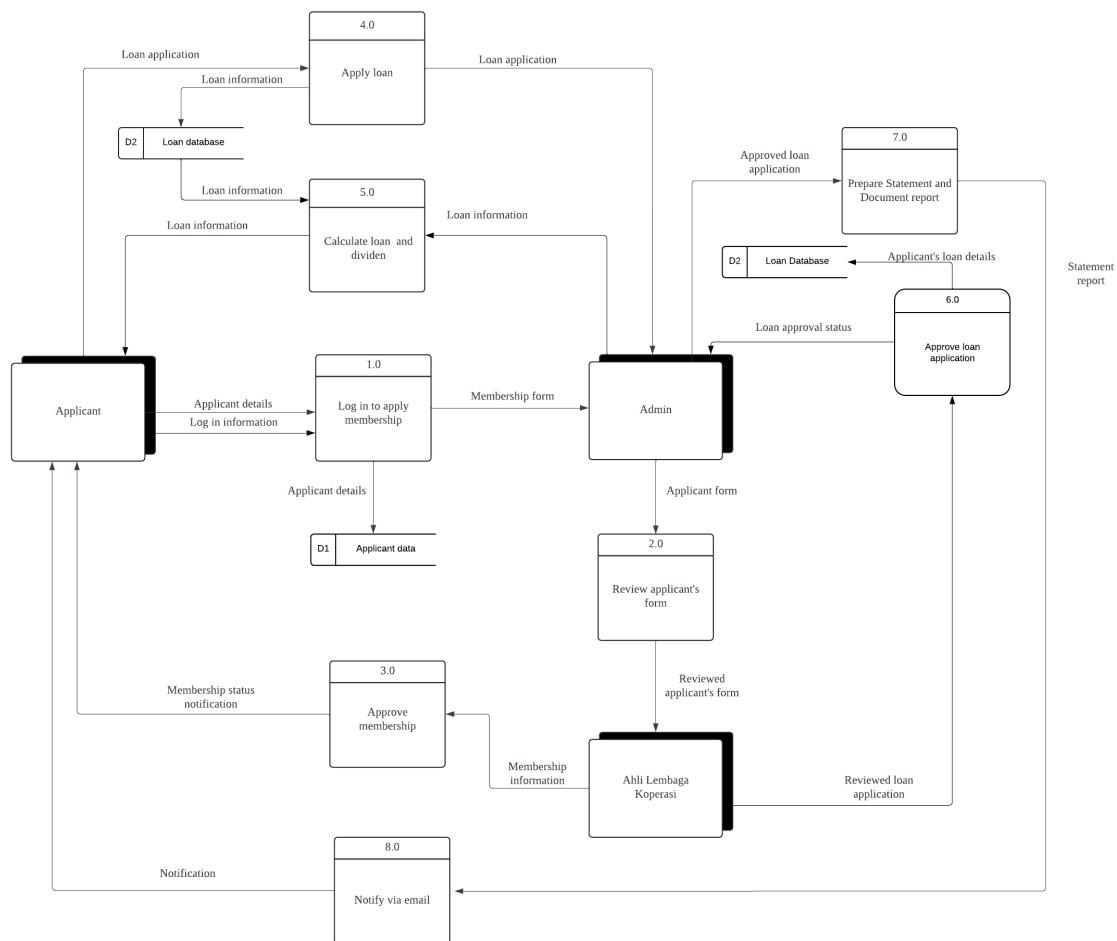


Figure 2: Level 0 Diagram

LOGICAL LEVEL 1 DIAGRAM

PROCESS 1 - Log in to apply membership

The process of logging in to apply for membership consists of five processes in Figure 3. The process starts with the applicant entering their credentials and submitting them in process 1.2. This information is then saved in the data store of applicant data (D1). The applicant uses their login credentials to access the system in process 1.3 after submitting. Process 1.4 is where the applicant fills out the membership form with their information after logging in. Lastly, in process 1.5, the applicant submits the completed membership form.

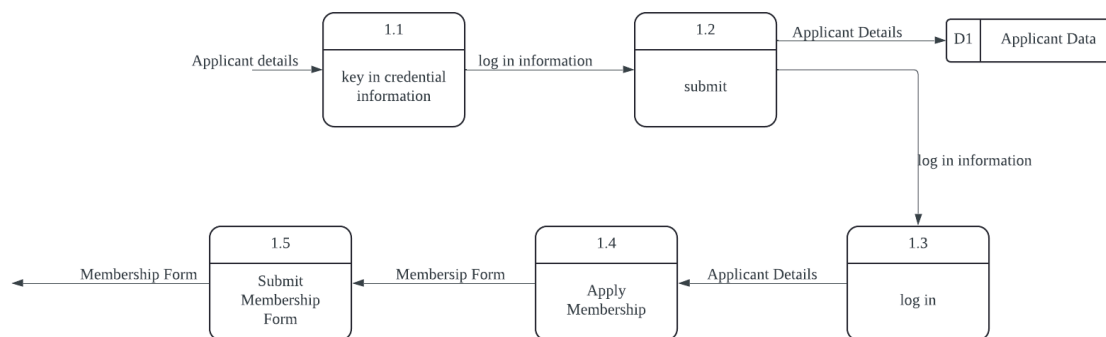


Figure 3: Logical Child Diagram for Process 1

PROCESS 2 - Review applicant's form

Figure 4 shows the process to review an applicant's form has three sub-processes. The applicant's form completion is first reviewed in process 2.1 to make sure all necessary fields are filled out. Next, the completed form proceeds to process 2.2, where it is reviewed for any errors. Process 2.3 involves validating the information entered to guarantee that all the data is true and valid. These processes produce a thoroughly reviewed applicant's form.

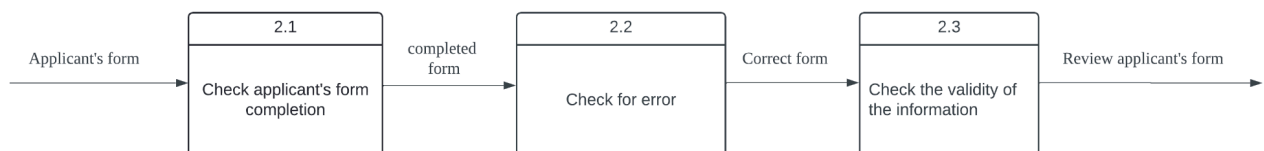


Figure 4: Logical Child Diagram for Process 2

PROCESS 3 - Approval membership

Figure 5 shows the process for handling a membership application. It begins with the submission of a membership form. In step 3.1, the applicant's details are verified to ensure all necessary information is provided. Once verified, step 3.2 involves validating the membership information to confirm its accuracy and completeness. In step 3.3, a decision is made to either approve or reject the membership application. If approved, the membership status is updated in step 3.4. Finally, in step 3.5, the applicant is notified of the decision regarding their membership status.

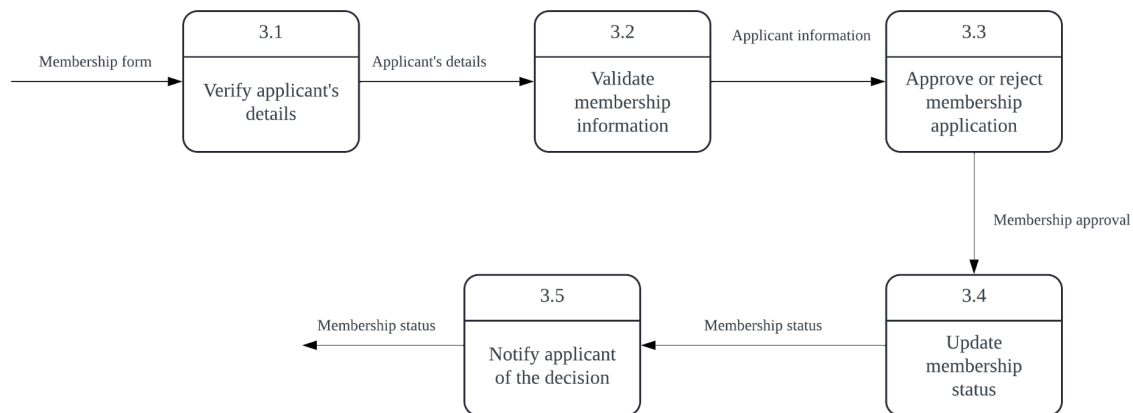


Figure 5: Logical Child Diagram for Process 3

PROCESS 4 - Apply loan

Figure 6 outlines the process for handling a loan application. It starts with the submission of a loan application form, which is then verified in step 4.1 to ensure all necessary details are provided. Once verified, the applicant's details are used in step 4.2 to assess their loan eligibility based on predefined criteria. If the applicant meets the eligibility requirements, the application proceeds to step 4.3, where a decision is made to either approve or reject the loan. If approved, the loan approval process is completed, resulting in the final output of loan approval.

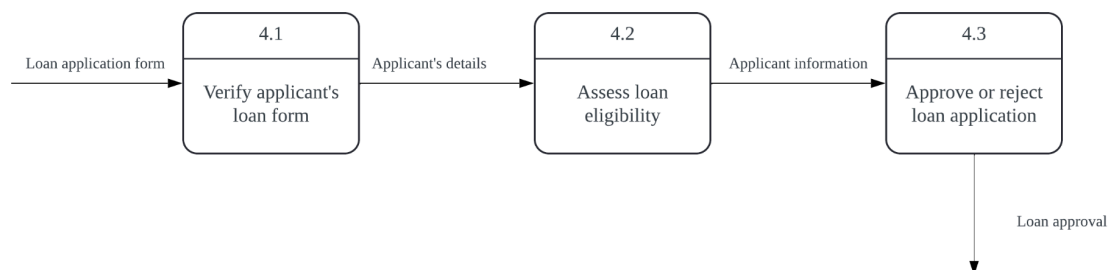


Figure 6: Logical Child Diagram for Process 4

PROCESS 5 - Calculate loan and dividend

Figure 7 shows the details of the sub-processes involved in calculating the loan and dividend. The process starts with retrieving loan application data 5.1 from the Loan Database. This data is then used to calculate the monthly loan repayment 5.2. Simultaneously, the system calculates the dividends based on the loan data 5.3. Finally, the results from the loan repayment and dividend calculations are presented to the applicant 5.4. The data flows include the loan application data, retrieved loan data, loan repayment details, loan information and dividend information, which move between the sub-processes to produce the final output.

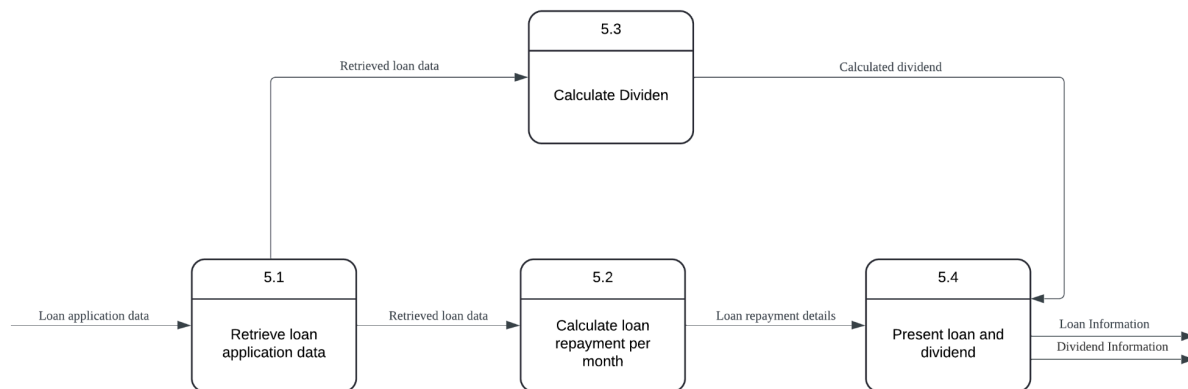


Figure 7: Logical Child Diagram for Process 5

PROCESS 6 - Approve loan application

Figure 8 shows the loan application approval process which consists of three primary steps. First, the loan application is reviewed in process 6.1, where the terms and conditions of the loan are evaluated. Next, in process 6.2, a decision is made regarding the approval or rejection of the loan based on the reviewed information. Finally, process 6.3 updates the loan records in the Loan Database (D2) with the applicant's details and the loan application status.

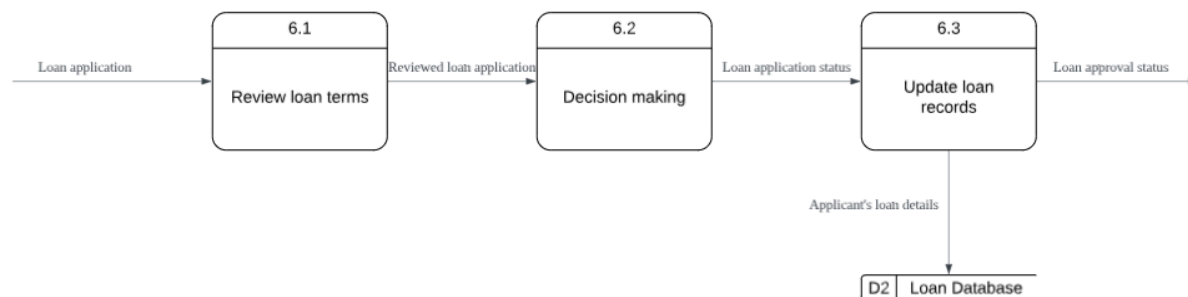


Figure 8: Logical Child Diagram for Process 6

PROCESS 7 - Prepare statement and document report

Figure 9 shows the process of preparing a report statement. First, the process begins with the approved loan application as input, leading to process 7.1 where applicant loan data is retrieved. This data is passed to process 7.2 to retrieve the corresponding loan repayments. Finally, process 7.3 generates the report statement using the collected loan data and repayment information.

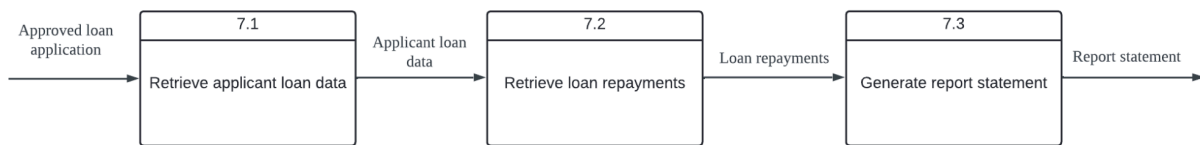


Figure 9: Logical Child Diagram for Process 7

PROCESS 8 - Notify via email

Figure 10 shows the process of sending loan approval notifications via email. It starts by fetching the notification details from a statement report (Step 8.1), then formats these details into a clear email (Step 8.2), and finally sends the email to the applicant (Step 8.3). This ensures applicants receive prompt and accurate loan approval notifications.

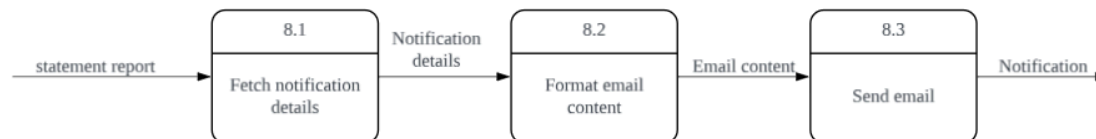


Figure 10: Logical Child Diagram for Process 8

Physical DFD

In Figure 11, the system for online membership and loan application is shown. The application process starts with the applicant logging into the system to apply for membership as shown in process 1.0. The administrator then reviews the applicant's form in process 2.0 and it will be approved by Ahli Lembaga Koperasi in process 3.0. Following approval of membership, the applicant may fill out an online loan application in process 4.0, performing both the loan and dividend estimation calculations automatically in process 5.0. After an admin approves the loan application in process 6.0, statements and reports are prepared as in process 7.0. Lastly, an email informing the applicant of their loan status is sent as shown in process 8.0. The diagram includes three data stores which are applicant data, membership lists, and loan databases.

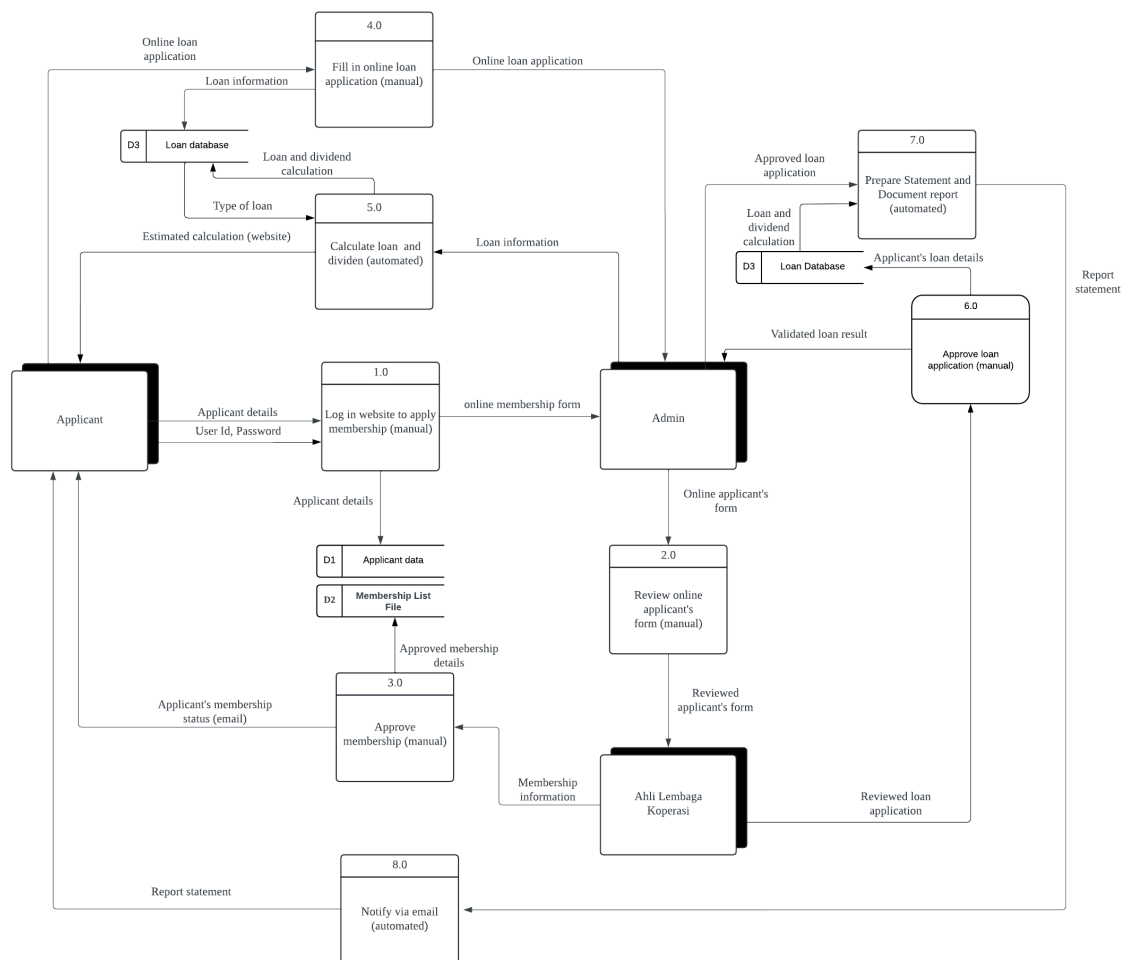


Figure 11: Physical DFD

PHYSICAL LEVEL 1 DIAGRAM

PROCESS 1 - Log in website to apply for membership (manual)

Figure 12 describes the login process to apply for membership. Applicants input their credential data into the login system interface in process 1.1. Submitting this data in process 1.2 creates user credentials, which consist of a password and user ID. The applicant next uses the credentials they were given to log into the website in process 1.3. After logging in successfully, applicants fill out the membership form in process 1.4. In process 1.5, they complete the membership form and submit it after reviewing all of the data that has been entered.

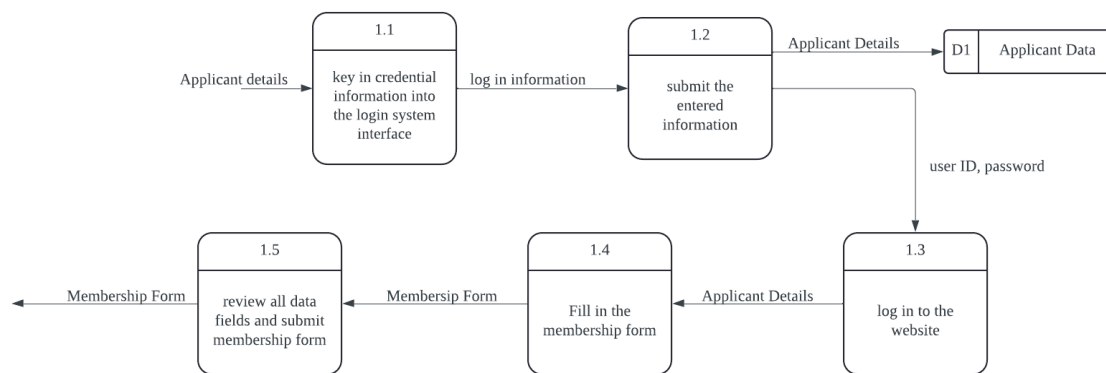


Figure 12: Physical Child Diagram for Process 1

PROCESS 2 - Review online applicant's form (manual)

The process of reviewing online applicant's forms is broken down into three processes in Figure 13. The first process is to ensure that the applicant has filled out the form completely and that all required information has been submitted. Process 2.2 is to check the form to make sure that all the information is correct. This is to ensure there are no errors and inconsistencies in the form. In order to make sure the application is based on accurate and trustworthy information, the last step which is process 2.3 is to validate and confirm the accuracy of the data submitted. The output for this child diagram is reviewed by the applicant's form.

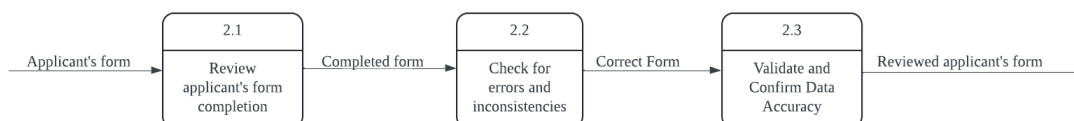


Figure 13: Physical Child Diagram for Process 2

PROCESS 3 - Approve membership (manual)

Figure 14 shows the process of approving membership applications. It starts with receiving the membership approval request and verifying the applicant's details to ensure all required information is correct. Once verified, the system updates the membership list with the new approved member's information and stores it in the Membership List File (D2). The final step involves sending an email notification to the applicant to inform them of their successful membership approval.

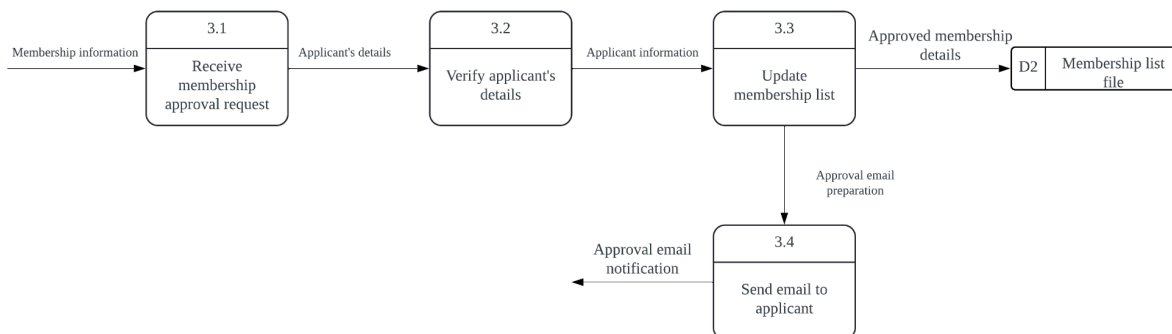


Figure 14: Physical Child Diagram for Process 3

PROCESS 4 - Fill in online loan application (manual)

Figure 15 illustrates the process of filling in an online loan application. The process begins with the applicant accessing the loan application form online. They then fill out the form with their personal and loan details. After completing the form, the applicant submits it, and the system validates the completeness of the information provided. Finally, the system records the submitted application in the Loan Database (D3), updating it with the applicant's details.

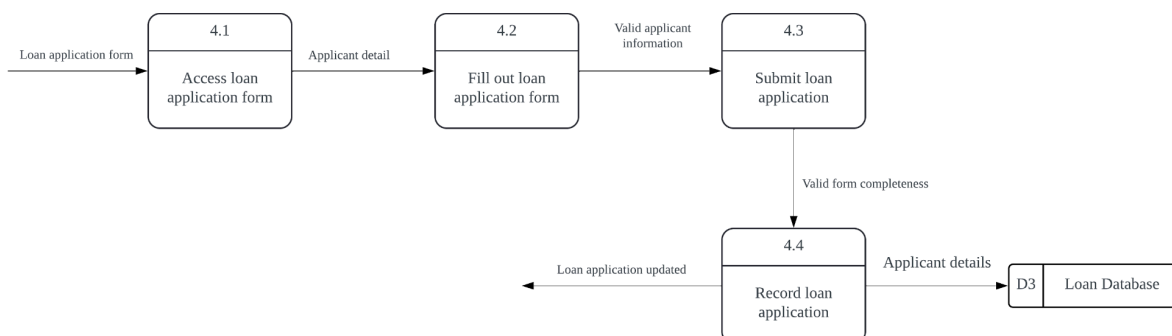


Figure 15: Physical Child Diagram for Process 4

PROCESS 5 - Calculated loan and dividend (automated)

Figure 16 demonstrates the workflow for calculating loan amounts within a system. It begins with the retrieval of loan application data 5.1 from a Loan Database (D3) based on a calculation request. The loan type is recorded and passed to two subsequent automated processes which are calculating the monthly loan repayment 5.2 and calculating the dividend 5.3. The calculated loan repayment and dividend are then sent to the website interface 5.4, which presents these calculations to the user.

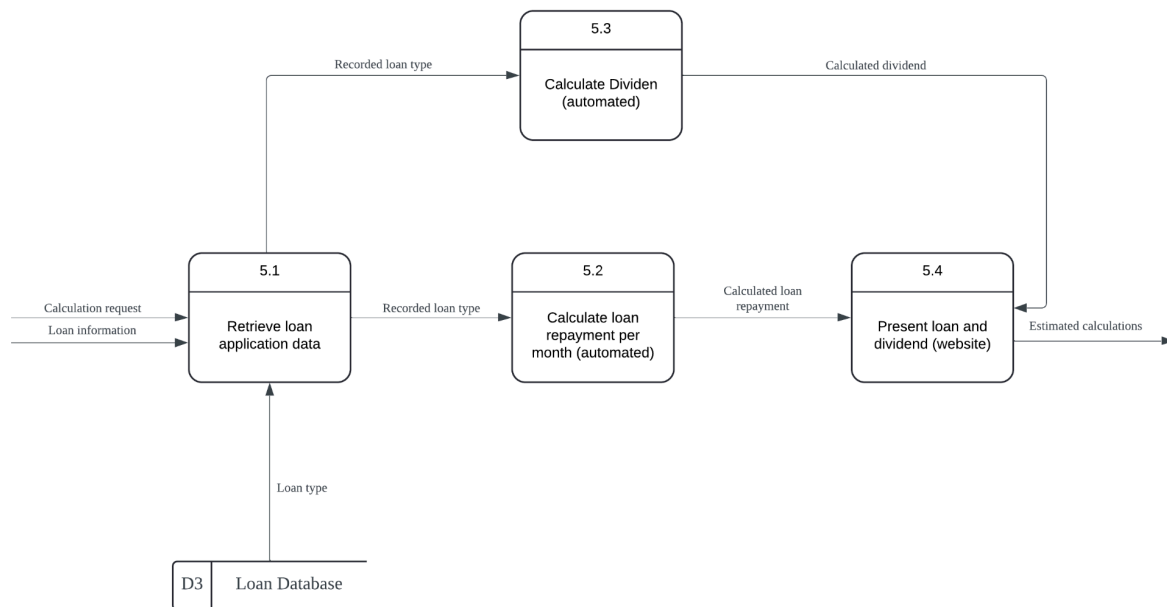


Figure 16: Physical Child Diagram for Process 5

PROCESS 6 - Approve loan application (manual)

Figure 17 illustrates the process of approving a loan application through three main steps. Initially, in the process 6.1, the loan application is reviewed against the loan terms obtained from the Loan Terms database (D1). Next, in process 6.2, the application is manually assessed for validity based on the reviewed information. Lastly, in process 6.3, the loan application status, along with the applicant's details and membership ID, is updated in the Loan Database (D2). This ensures that each loan application is meticulously reviewed, validated, and recorded.

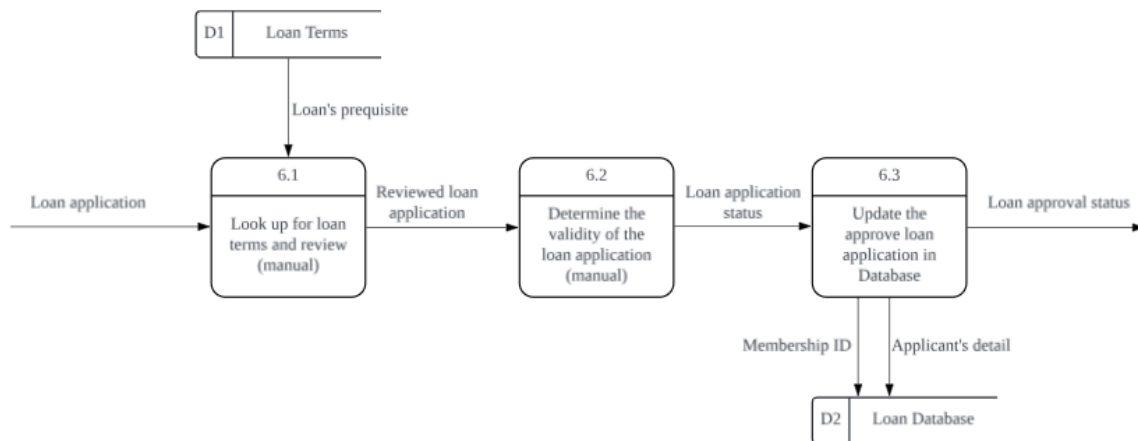


Figure 17: Physical Child Diagram for Process 6

PROCESS 7 - Prepare statement and document report (automated)

Figure 18 shows the process of preparing a report statement. This process begins with the approved loan application as input, leading to process 7.1 where applicant loan data is retrieved. Applicant loan details are passed to process 7.2 to retrieve the corresponding loan repayments automatically from the loan database (D2). Finally, process 7.3 generates the report statement using the collected loan data and repayment information.

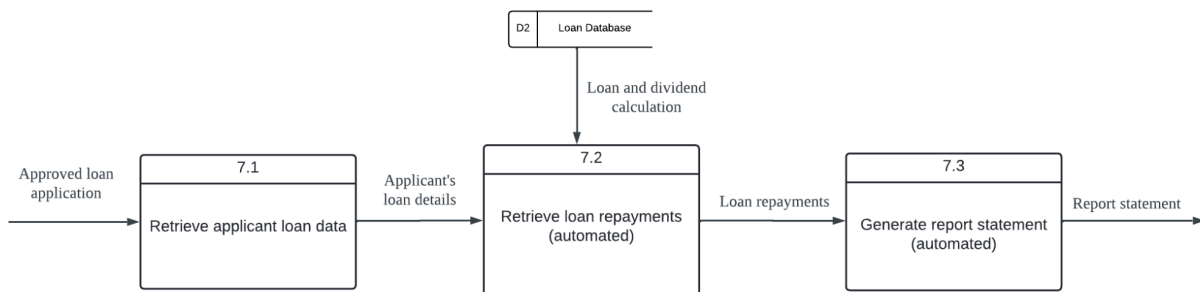


Figure 18: Physical Child Diagram for Process 7

PROCESS 8 - Notify via email (automated)

Figure 19 illustrates the process of notifying applicants via email. This process starts with process 8.1, which fetches the notification details from the statement report and retrieves the applicant's email address. This information is then used by process 8.2 to automatically write the email messages. Once the email messages are prepared, they are passed to process 8.3, which sends the emails automatically to the applicants. The final output is the report statement being delivered to the applicants' email addresses.

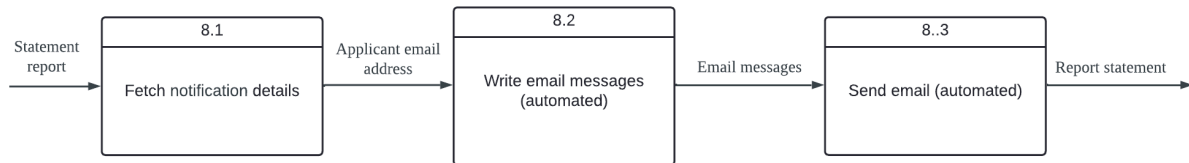


Figure 19: Physical Child Diagram for Process 8

CRUD

SCENARIO 1 : MEMBERSHIP APPLICATION

Table 1 illustrates operations performed by various entities (Applicant, Admin, System) during the membership application and approval processes. The table specifies that the applicant logs in and applies for membership, the system handles the form submission and notifies the applicant upon successful submission, and the admin retrieves and reviews the membership form.

Entity	Applicant	Admin	System	Applicant Dta (D1)	Notification
Applicant logs in to apply for membership	C, R		C	C	
Applicant submits membership form	C		R	U	
Admin reviews membership form	R			R	

Table 1: CRUD Table Scenario 1

SCENARIO 2 : MEMBERSHIP APPROVAL

Table 2 illustrates the interactions between different entities and data operations within a membership approval process. When the admin approves membership, both the admin's records and the applicant's data are updated. Following this, the system generates an approval notification. Finally, the admin reads the notification, and the applicant receives and reads the membership approval email.

Entity	Admin	Applicant Data (D1)	Notification	ALK
Admin approves membership	U	U		
System generates approval notification			C	
Applicant receives membership approval email	R		R	
Admin sent reviewed membership form to ALK				R, U
ALK approve membership and loan				U

Table 2: CRUD Table Scenario 2

SCENARIO 3 : LOAN APPLICATION

Table 3 outlines the interactions between different entities (Applicant, Admin, System) and the loan database during the loan application process. When the applicant applies for a loan, they create a new entry, and the system also creates this entry in the loan database. The system then calculates the loan and dividend, updating its records and the loan database accordingly. Finally, the system stores the calculated loan and dividend information, with the admin updating the records in the loan database.

Entity	Applicant	Admin	System	Loan Database (D2)
Applicant applies for a loan	C		C	C
System calculates loan and dividend			C, U	U
System stores calculated loan and dividend			U	U

Table 3: CRUD Table Scenario 3

SCENARIO 4 : LOAN APPROVAL

Table 4 outlines the data operations performed by different entities (Admin, System) on the loan database and notifications. The admin retrieves approved loan applicant data by reading from both the system and the loan database. The system then prepares the statement report, creating the report and updating the loan database with the new information. Finally, the system sends a notification, creating an email to inform the applicant that their statement report is ready.

Entity	ALK	Loan Terms (D1)	Loan Database (D2)
ALK receives and reviews loan application	R	R	
ALK approve loan application			C
ALK updates applicants name into the system			U

Table 4: CRUD Table Scenario 4

SCENARIO 5: STATEMENT REPORT

Table 5 details the interactions between different entities (Admin, System, Loan Database, and Notification) in the process of handling a statement report. The admin retrieves approved loan applicant data from both the system and the loan database involving read operations. The system then prepares the statement report, involving a create operation on the system and an update operation on the loan database (D2). Finally, the applicant receives the statement report via email, which triggers a create operation in the notification system.

Entity	Admin	System	Loan Database (D2)	Notification
Admin retrieve approved loan applicant data	R		R	
System prepares statement report		C	U	
Applicant receives statement report via email				C

Table 5: CRUD Table Scenario 5

EVENT RESPONSE TABLE

SCENARIO 1 : MEMBERSHIP APPLICATION

Table 6 outlines the process of a membership application through three main events: logging in, submitting the membership form, and reviewing the form. Each event is broken down into its source (Applicant or Admin), the trigger for the event, the specific activity involved, the system's response, and the destination where the data is stored. For instance, when an applicant logs in, their details are validated and stored in the system as Applicant Data (D1). Upon submitting the membership form, the form data is validated and stored in both Applicant Data (D1) and the Membership List File (D2). Finally, when the admin reviews the form, they check for completeness, validate the details, and update the review status in the system, with the information being reflected in the Applicant Data (D1).

Event	Source	Trigger	Activity	Response	Destination
Applicant logs in to apply for membership	Applicant	Applicant provides login details	Validate login and store applicant details	Applicant details stored in the system	Applicant Data (D1)
Applicant submits membership form	Applicant	Applicant fills and submits membership form	Validate and store form data in the Applicant Data (D1)	Membership form data is stored	Membership list File (D2)
Admin reviews membership form	Admin	Admin accesses the membership form	Check for completeness and validate form details	Review status updated in the system	Applicant Data (D1)

Table 6: Event Response Table Scenario 6

SCENARIO 2 : MEMBERSHIP APPROVAL

Table 7 outlines the steps involved in approving a membership application. Initially, the Admin approves the membership, triggered by their decision to approve the application. This action updates the membership status to "approved" in Applicant Data (D1). Following this, the system generates an approval notification email, triggered by the status update. The activity involves creating and sending this email to the applicant, resulting in the applicant receiving the notification. The final step occurs when the applicant receives the approval email in their inbox, triggered by the email's arrival. The applicant then opens and reads the notification, informing

them of their membership approval. This table details each event's source, trigger, activity, response, and destination, highlighting the flow from membership approval to the applicant being informed.

Event	Source	Trigger	Activity	Response	Destination
Admin approves membership	Admin	Admin decides on membership approval	Update membership status and save in Applicant Data (D1)	Membership status is updated to approved	Applicant Data (D1)
System generates approval notification	System	Membership status is updated to approved	Generate approval notification email and send it to applicant	Approval notification email is sent to applicant	Applicant
Applicant receives membership approval email	Applicant	Email received in applicant's inbox	Open and read approval notification	Applicant is informed of membership approval	Applicant
Admin sent reviewed membership form to ALK	Admin	Membership approval requires ALK review	Send reviewed membership to ALK	ALK receives the reviewed membership form	ALK
ALK approves membership and loan	ALK	ALK receives reviewed membership form	Approve membership and loan	Membership and loan are approved	Applicant Data (D1)

Table 7: Event Response Table Scenario 2

SCENARIO 3 : LOAN APPLICATION

Table 8 illustrates a detailed sequence of interactions between the applicant and the system. It begins with the event where the applicant applies for a loan. The source is the applicant, and the trigger is the submission of the loan application. The system's activity is to validate the loan application details and store the application in the Loan Database (D2), with the response being the stored loan application. Next, the system calculates the loan and dividend once the application is stored, triggered by the stored loan application. The activity here is the calculation process, and the response is the completion of these calculations, which are again

stored in the Loan Database (D2). The final event involves the system storing the calculated loan and dividend, triggered by the completion of calculations, with the activity being the actual storage process and the response confirming the storage in the Loan Database (D2).

Event	Source	Trigger	Activity	Response	Destination
Applicant applies for a loan	Applicant	Applicant submits loan application	Validate loan application details and store loan application in the Loan Database (D2)	Loan application is stored	Loan Database (D2)
System calculates loan and dividend	System	Loan application is stored	Calculate loan and dividend	Loan and dividend calculations are completed	Loan Database (D2)
System stores calculated loan and dividend	System	Loan and dividend calculations are completed	Store calculated loan and dividend	Calculated loan and dividend are stored	Loan Database (D2)

Table 8: Event Response Table Scenario 3

SCENARIO 4 : LOAN APPROVAL

Table 9 provides a detailed overview of the steps involved in the loan approval process. The first event is ALK receiving the loan application from the Admin, triggered when the loan application is ready for ALK review. ALK reads the loan terms to review the application, resulting in a reviewed loan application stored in Loan Terms (D1). The second event is ALK approving the loan application, triggered by the review of the application. The activity involves ALK approving the application, with the response being an approved loan application stored in the Loan Database (D2). The final event is ALK updating the applicant's name in the system, triggered by the determination of a valid loan application. ALK updates the applicant's details in the database, resulting in updated applicant details stored in the Loan Database (D2).

Event	Source	Trigger	Activity	Response	Destination
ALK receives and review loan application	Admin	Loan application is ready for ALK review	ALK reads loan terms to review the application	Reviewed loan application	Loan Terms (D1)
ALK approve loan application	ALK	Loan application is reviewed by ALK	ALK approves the application	Approved loan application	Loan Database(D2)
ALK updates applicants name into the system	ALK	Valid loan application determined	ALK updates applicant's details in the database	Updated applicant's details	Loan Database(D2)

Table 9: Event Response Table Scenario 4

SCENARIO 5 : STATEMENT REPORT

Table 10 outlines the sequence of events involved in generating and delivering a statement report to a loan applicant. The process begins with the admin retrieving approved loan applicant data from the system, triggered by the availability of an approved loan application list. The activity involves retrieving the data, and the response is to prepare the statement report, with the destination being the system. Next, the system prepares the statement report when the loan status is updated to approved. System generates and sends the report via email to the applicant, with the response being the generation and sending of the statement report. Finally, the applicant receives the statement report via email upon receiving a notification, applicant opens the notification, reads the statement report details, and is informed of the loan approval and calculation.

Event	Source	Trigger	Activity	Response	Destination
Admin retrieve approved loan applicant data	Admin	Approved loan application list	Retrive the approved loan applicant data	Prepare statement report	System
System prepares statement report	System	Loan status is updated to be	Generate statement	Statement report	Applicant

		approved	report and send it via email	generated and sent	
Applicant receives statement report via email	Applicant	Notification received	Open notification and read statement report details	Applicant is informed of loan approval and calculation	Applicant

Table 10: Event Response Table Scenario 5

Figure 20 breaks down the Loan Application System into three main modules: Membership Management, Loan Processing, and Notification System. The Membership Management module handles member applications, verification, approval, and management of membership details through four submodules. The Loan Processing module manages loan applications, including filling out forms, calculating loans, verifying eligibility, approving loans, and managing the loan database through five submodules. The Notification System module sends email notifications about loan statuses and updates. Data flows between these modules, starting with the application form, then updating membership and loan data, and finally notifying applicants of their loan status. This chart clearly shows how each part of the system works together to process loans and manage memberships efficiently.



PARTITIONING

Process 1

This is the initial step where the applicant logs in to the website to submit an online membership application form. This process involves the applicant providing their user ID and password and filling out the membership form. This process was partitioned individually due to different users being involved in each step and it does not occur simultaneously with other processes.

Process 2

After the applicant submits the membership form, the form is reviewed by the admin. This step involves a manual review of the submitted details to verify the applicant's information. This process was partitioned individually due to different users being involved in each step and it does not occur simultaneously with other processes.

Process 3

Every application is assessed uniformly based on established criteria by partitioning the membership approval process. Ahli Lembaga Koperasi are able to carefully assess candidates' backgrounds and records to make sure they fit the requirements of the organization. This error by humans helps maintain the consistency and keeping the high standards in the membership selection process.

Process 4-5

Real-time computations are possible when these procedures are combined. With the ability to calculate loan terms and possible dividends instantaneously as applicants enter their loan details, the system can give applicants instant feedback. This enhances decision-making and user experience.

Process 6-7

Similar kinds of data, including loan details and customer information, are accessed and updated during both processes. A more efficient data flow becomes possible by integrating these operations. The information needed to create statements and reports is instantly available within the same workflow after a loan application is approved. Since the data required to prepare statements comes directly from the loan approval process, combining these processes also helps to verify that the data is correct and current. This lowers the possibility of errors and increases document quality.

Process 8

Handling sensitive client data is a common aspect of email communication. This process can be partitioned so that we can apply email-specific compliance checks and particular security

measures. This guarantees that client data is properly secured and that we comply with regulatory standards

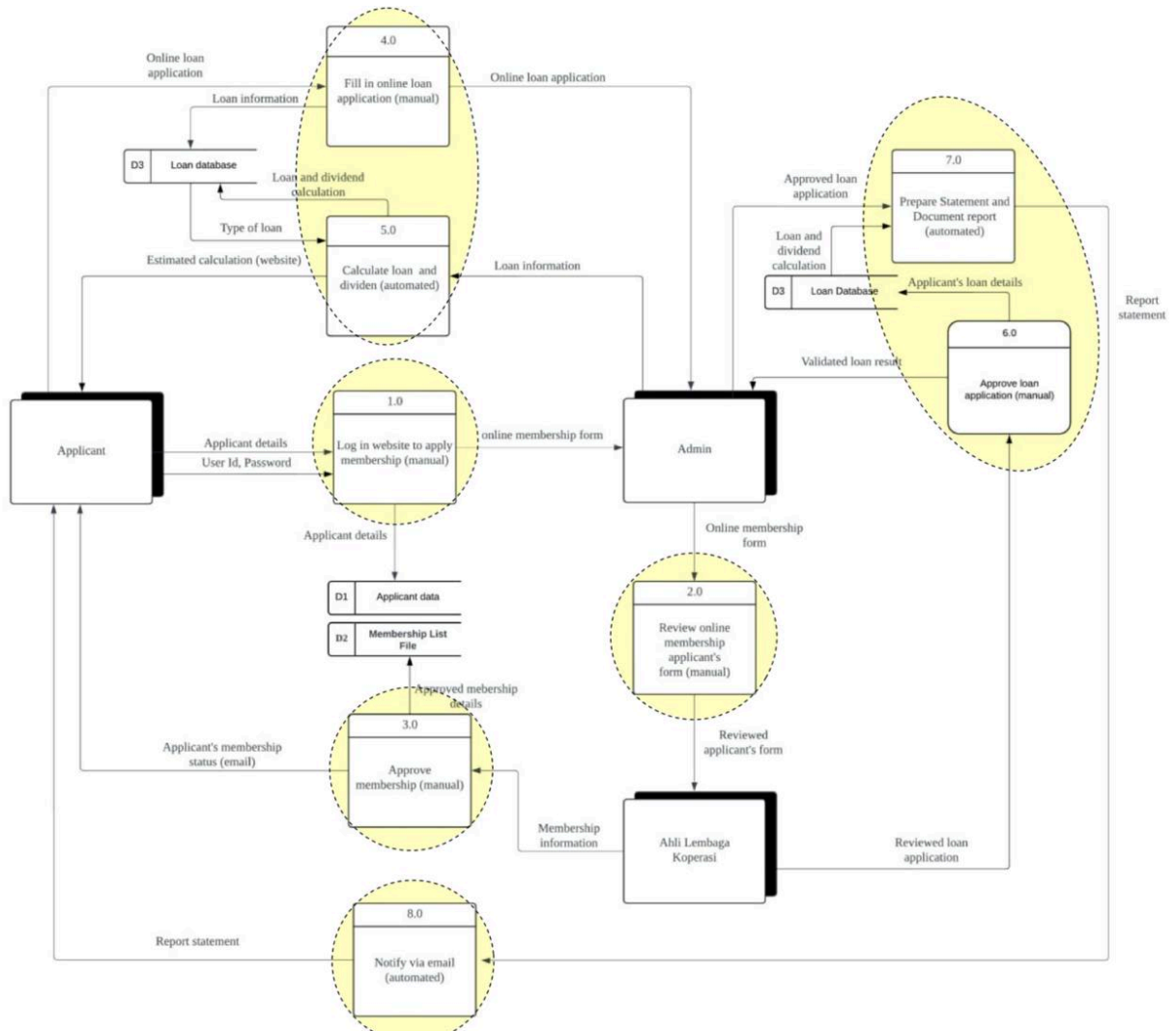


Figure 21: Structure Chart

SYSTEM ARCHITECTURE

Based on the physical DFD and system architecture generated, the Koperasi system can be implemented via a web-based approach. A web-based system for the Koperasi is ideal because of its accessibility, allowing users (applicants and administrators) to access the system from anywhere using a web browser without the need to install any software. It also enables real-time processing and fast feedback on application status. The system's three-layer architecture in Figure 22 ensures effective networking and communication. The presentation layer is made up of web browsers on users' computers, tablets, and smartphones; the business layer is made up of web servers hosting the application logic; and the data layer is made up of database servers managing data storage and retrieval.

Detailed Descriptions:

1. User Interface (UI)

Web application that allows applicants to apply for membership and loans and for administrators to review and approve these applications.

2. Authentication

Manages user login credentials and controls access to different parts of the system based on user activity.

3. Membership Application

Allows applicants to fill in and submit online membership forms.

4. Loan Application

1. Allows members to fill in and submit loan application forms.
2. Automates the loan and dividend calculation based on input data.
3. Automates the preparation of loan statements and reports.

5. Approval

1. Allows administrators to review and approve membership applications.
2. Admin approves membership applications manually.

6. Website Loan and Dividend Calculator

Automates calculations related to loans and dividends based on the type of loan and other parameters provided by the user.

7. Notification

Sends email notifications to applicants about their application status and other updates.

8. Database

1. Applicant Data (D1): Stores details of applicants.
2. Membership List File (D2): Stores data of approved members.
3. Loan Database (D3): Stores loan application information and calculation results.

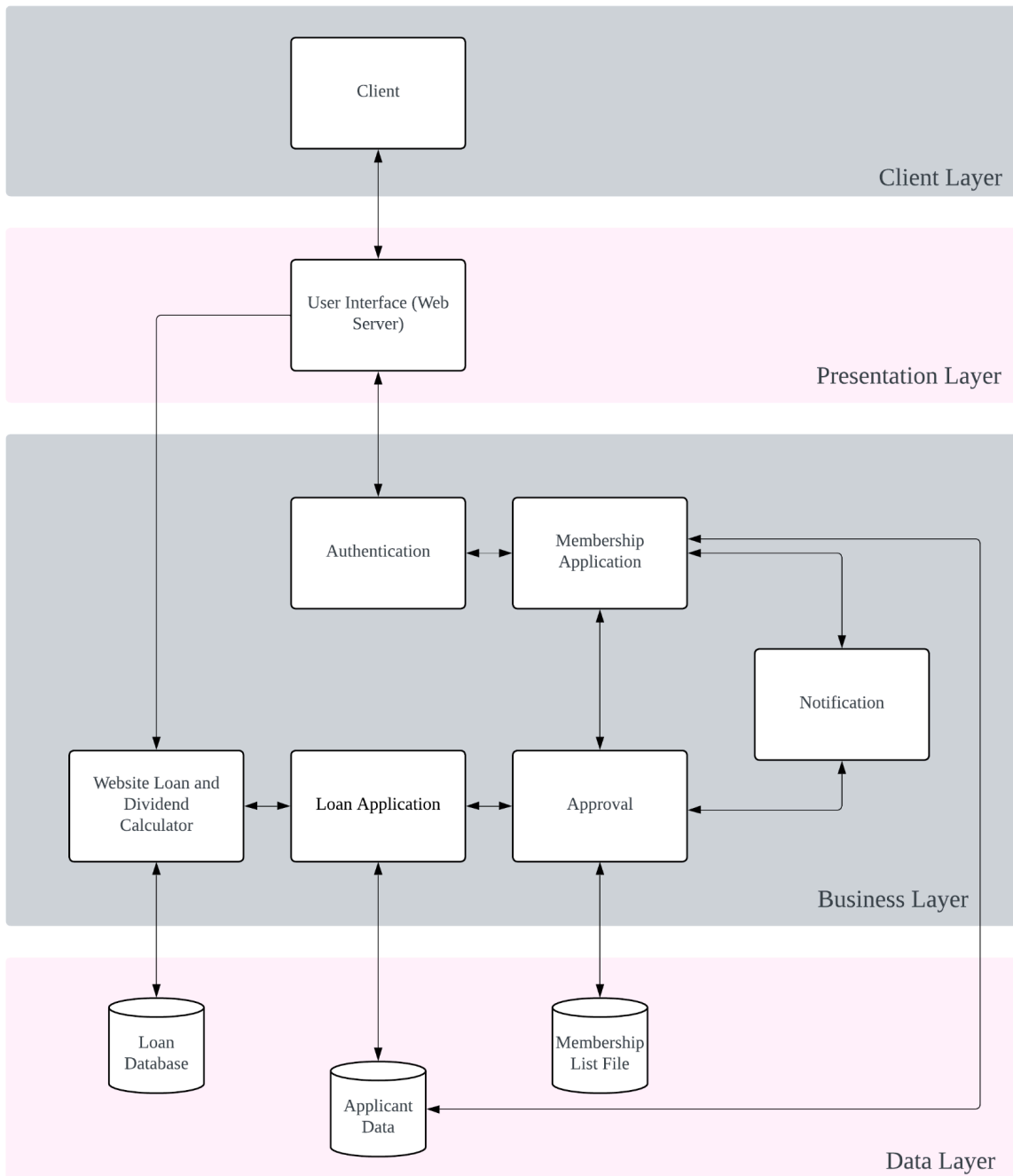


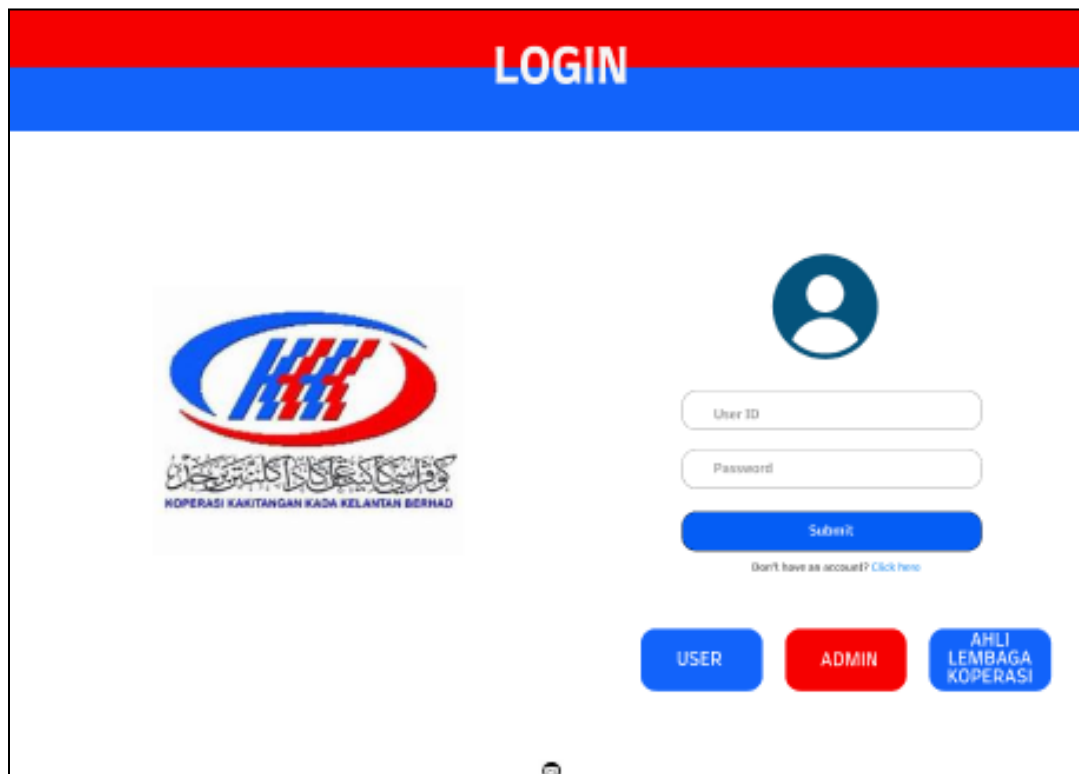
Figure 22: System Architecture

WIREFRAME

1. User Interface

1.1 User Login

When opening the Koperasi Kada app, Figure 23 shows that users must log in with their information. After logging in, Figure 24 allows users to choose from various menu options, including applying for membership, applying for a loan, and calculating loan details.



The wireframe for the User Login screen features a red header bar with the word "LOGIN" in white. Below the header is a blue horizontal bar. The main content area is white and contains the Koperasi Kada logo on the left, which includes a stylized "KK" in blue and red, and the text "KOPERASI KAKITANGAN KADA KELANTAN BERHAD" below it. On the right side, there is a user profile icon (a blue circle with a white person silhouette), followed by two input fields labeled "User ID" and "Password". Below these fields is a blue "Submit" button. Under the "Submit" button is a link that says "Don't have an account? Click here". At the bottom right, there are three buttons: "USER" (blue), "ADMIN" (red), and "AHLI LEMBAGA KOPERASI" (blue).

Figure 23: User Login

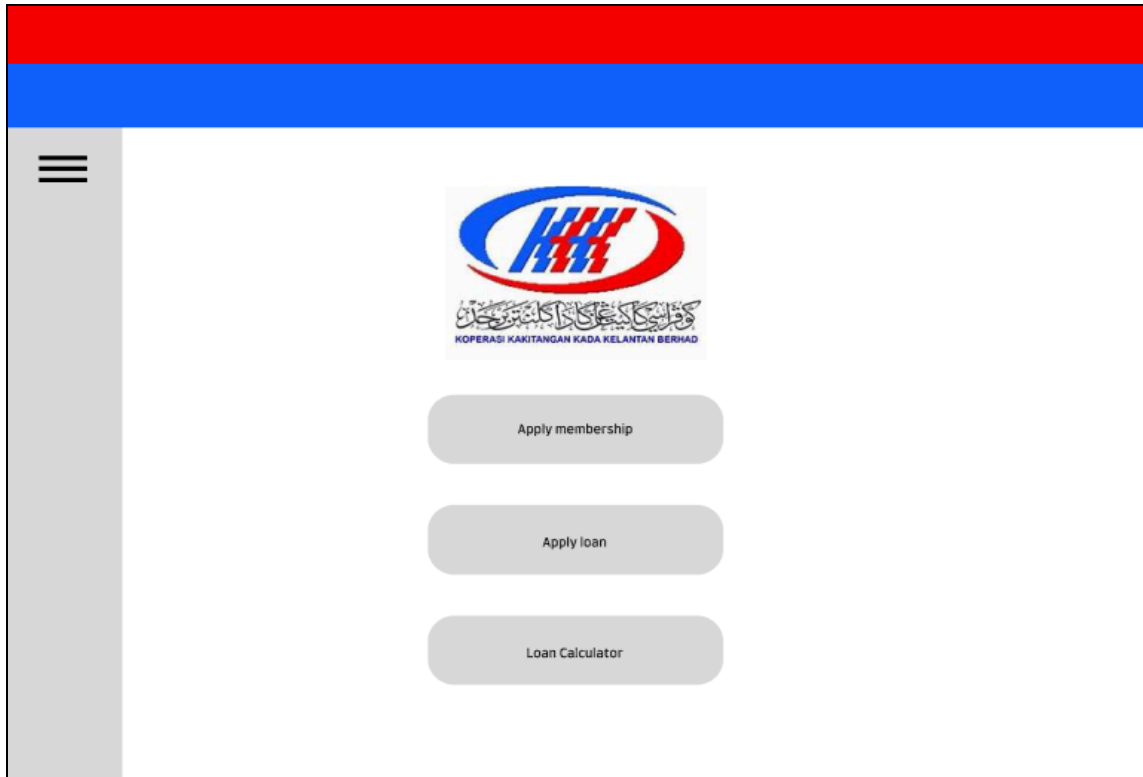
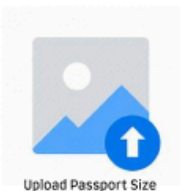


Figure 24: User Menu

1.2 Membership application

When users choose to apply for membership, they are directed to Figure 25, where they must fill out the membership form. After clicking the "Next" button, they proceed to Figure 26 to complete additional sections of the membership form. Once all required information is filled in and they click "Submit," Figure 27 displays their membership status.

MEMBERSHIP FORM

Upload Passport Size photo here

Applicant's Information

Name : _____	I/C Number : _____
Sex : Male Female Religion : _____	Marital Status : Married Single Divorced
Address : _____	Postcode : _____
	State : _____
Staff Number : _____ PF Number : _____	Position & Grade : _____
Office Address : _____	Postcode : _____ Fax/Tel Number : _____
	Phone Number : _____
House Phone Number : _____	Monthly Salary : _____

SAVE
NEXT

Figure 25: Membership form page 1

MEMBERSHIP FORM


Applicant's Family Information

Num	Relationship	Name	I/C Number

[Click here to add column](#)

Fees and Donations

Num	Activity	RM
1	Entrance Fee	
2	Share Modal*	
3	Fee Modal	
4	Membership Deposit	
5	Tabung Kebajikan Donation (AI-Abrar)	
6	Saving	
7	Others	

*Minimum share modal is RM300.00 and not exceeding 1/5 of Koperasi share modal and the payment must be made within 6 months from the date of membership approval

SAVE
SUBMIT

Figure 26: Membership form page 2

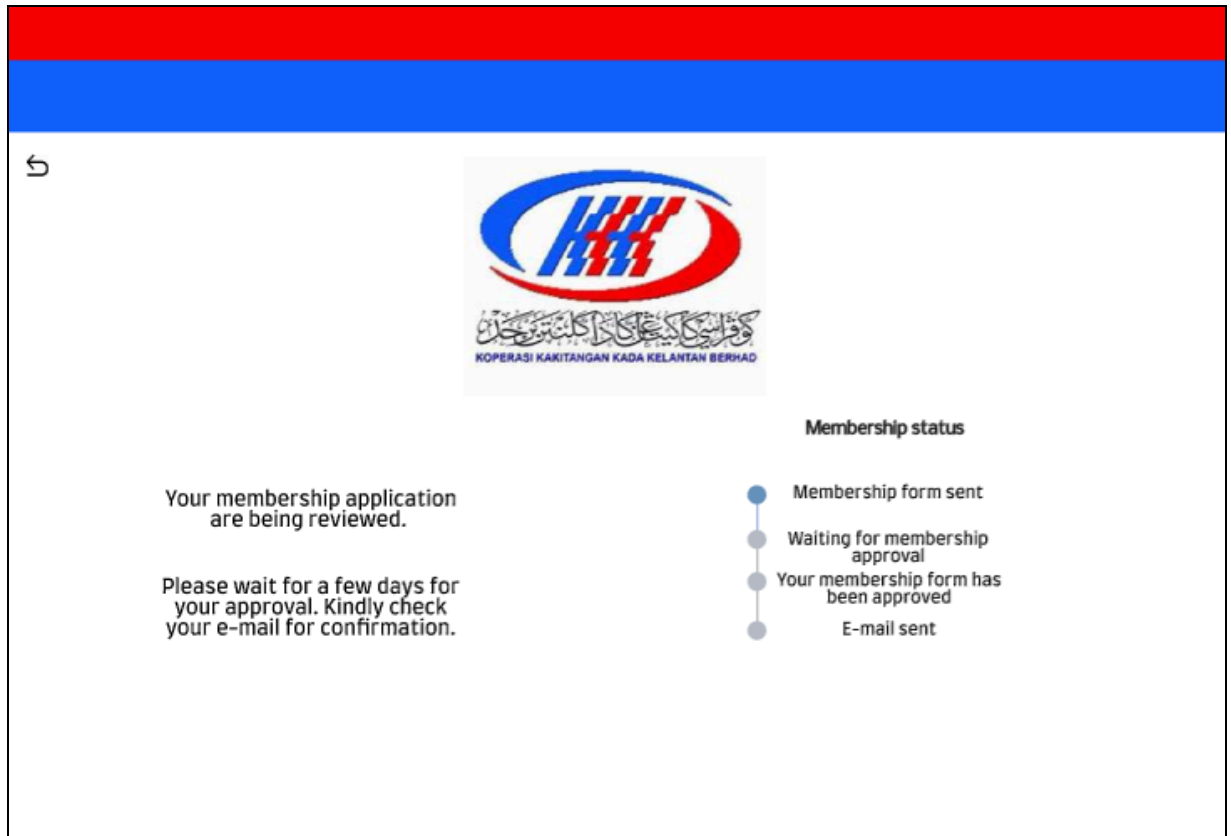


Figure 27: Membership status

1.3 Loan Application Process

A user must first log in as a member as shown in Figure 28 in order to apply for a loan using the website. Then, before filling out the loan application form as in Figure 30 with the loan type, amount, and duration, they can utilize the loan calculator to estimate monthly repayments in Figure 29. In the applicant details area, the user is also required to enter their financial and personal data as in Figure 31. In Figure 33 and Figure 34, the user must provide the details of a guarantor and obtain the guarantor's agreement to the terms. Lastly, the application status will be shown as shown in Figure 35, indicating whether the loan is pending approval, is being reviewed, or needs further information.

MEMBERSHIP LOGIN





[Don't have an account? Click here](#)

Figure 28: Membership Login



LOAN CALCULATOR

Loan Amount

RM

Loan Duration

Years

Interest Rate

%

Your Estimated Monthly Repayment

RM 0.00

Figure 29: Loan Calculator

LOAN APPLICATION FORM



[←](#)

LOAN DETAILS

Type of loan :	☐ Al - Bai	☐ Al - Inah	☐ Skim Khas
	☐ Baik Pulih Kenderaan	☐ Cukai Jalan	☐ Karnival Musim Istimewa
	☐ Lain- lain (Nyatakan) _____		

Loan Amount : RM ,

Loan Duration : month

Monthly Repayment : RM ,

SAVE
NEXT



LOAN APPLICATION FORM

APPLICANT LOAN DETAILS

I/C Num : 950122-10-0000

D.O.B : 22 / 01 / 1995

Age : 29 Years Old

Sex : F

Religion : ISLAM

Race : MELAYU

Address : NO 1 JALAN INAISARI 5, BANDAR SG LIMAU, 48000, RAWANG, SELANGOR.

Postcode : 48000

City : RAWANG

State : SELANGOR

Staff Number : A23000001

PF No. : 1113597

Position : Pengurus Kewangan, W54

Office Address : Ibu Pejabat Kada Lundang, 15710, Kota Bharu, Kelantan, Malaysia.

Postcode : 15710

City : Kota Bharu

Phone Number : 010-2223333

Bank Name/CWG : MBCI BANK

Account Bank Num. : 1000123458907654

[BACK](#)

[SAVE](#)

[NEXT](#)

LOAN APPLICATION FORM



APPLICANT AGREEMENT

I I/C Num. :

hereby authorize KOPERASI KAKITANGAN KADA BHD or its legal representative to obtain any necessary information and also to get a refund from my salary deduction and emoluments as the amount loaned. I also agree to accept any decision from this KOPERASI to reject the application without giving any reason

BACK
SAVE
NEXT



LOAN APPLICATION FORM

GUARANTOR DETAILS

Guarantor 1

NAME : _____

IC NUMBER : _____

PF NUM. : _____ ID NUM : _____

LATEST PAY SLIP : Upload File

Guarantor 2

NAME : _____

IC NUMBER : _____

PF NUM. : _____ ID NUM : _____

LATEST PAY SLIP : Upload File

BACK
SAVE
NEXT

LOAN APPLICATION FORM



GUARANTOR AGREEMENT

☐ With this agreement, I/we voluntarily agree, either jointly or individually, to accept full responsibility for any loans made to cover the cost of these items or finance. If the borrower cannot return the loans in accordance with the agreed upon terms, we guarantee to reimburse all of the debt payments that were issued to him above.

BACK

SAVE

SUBMIT

⤵



Loan status

- Loan application form sent
- Waiting for loan approval
- Your loan has been approved
- E-mail sent

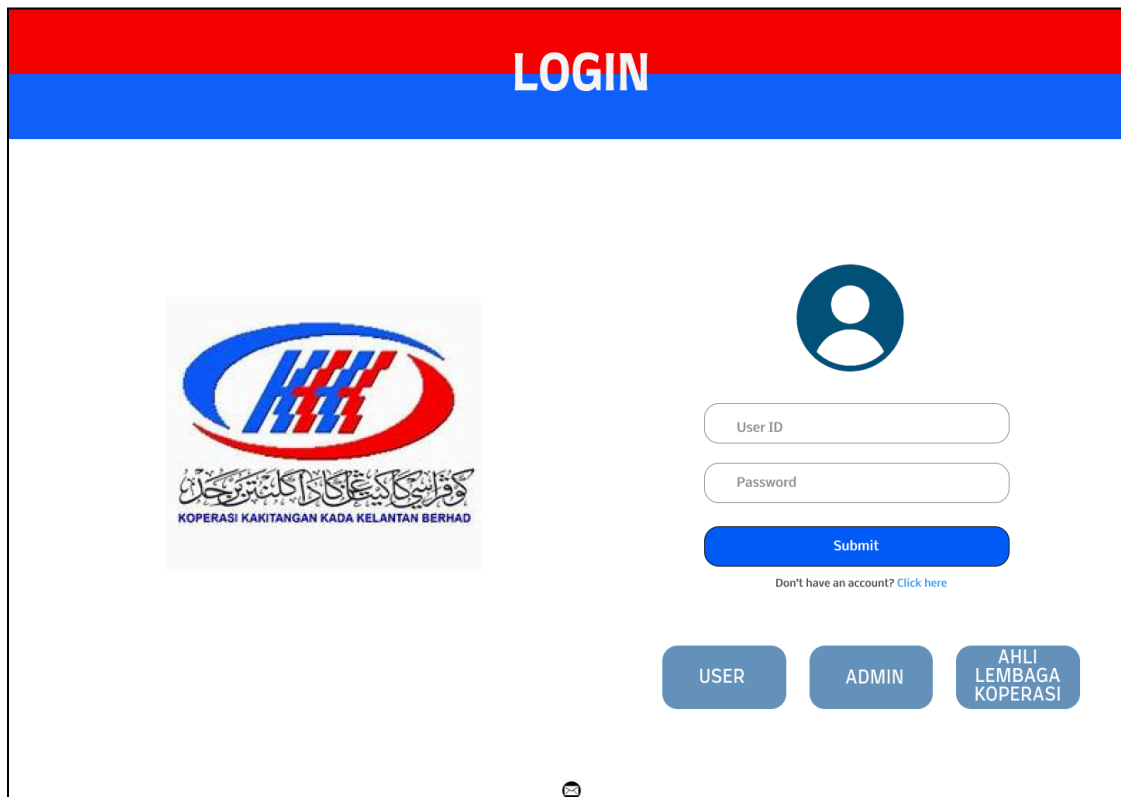
Your loan application are being reviewed.

Please wait for a few days for your approval. Kindly check your e-mail for confirmation.

2. Admin

2.1 Admin login

To log in as an admin, click the "ADMIN" button on the login page as shown in Figure 36. Enter your User ID and Password, then click "Submit" to access the admin features. After successfully logging in as an admin, you will see options to manage various aspects of the organization. These options include viewing the membership list, checking the loan application list, and viewing members approved by ALK as shown in Figure 37.



The login page features a red header with the word "LOGIN" in white. Below the header is a blue horizontal bar. The main content area is white and contains the logo of Koperasi Kakitangan Kada Kelantan Berhad on the left. On the right, there is a user profile icon, input fields for "User ID" and "Password", a blue "Submit" button, and a link "Don't have an account? Click here". At the bottom, there are three buttons: "USER", "ADMIN", and "AHLI LEMBAGA KOPERASI".

Figure 36: Login Page

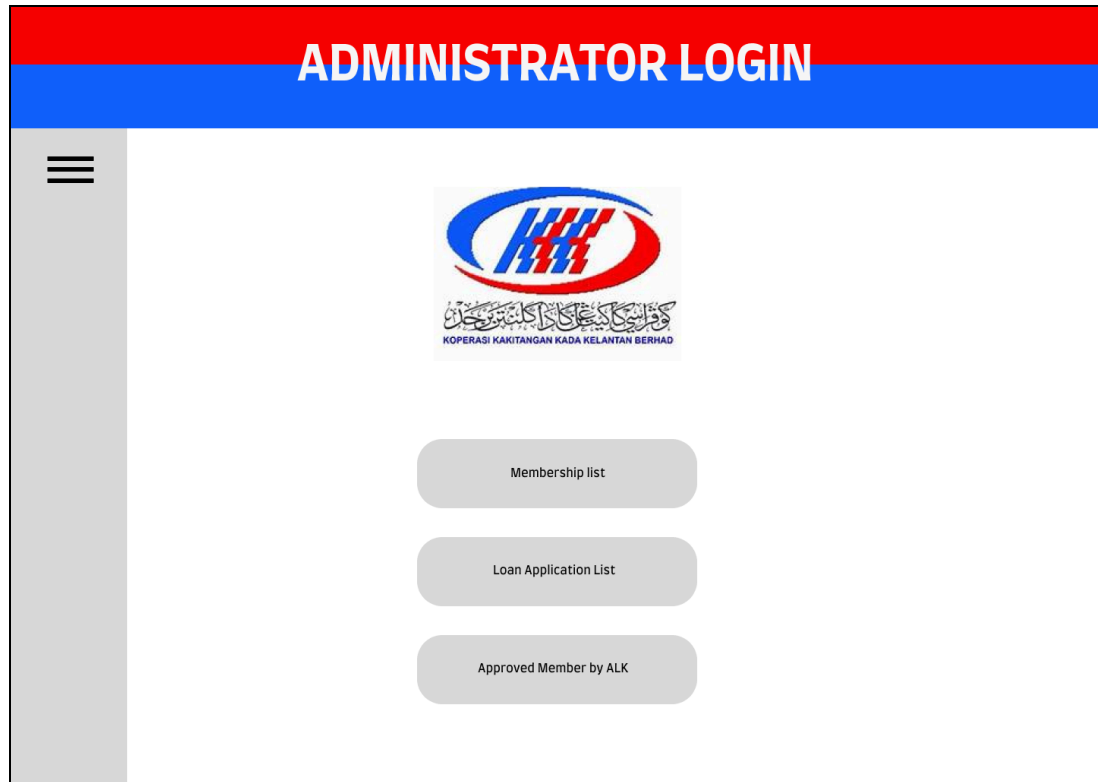


Figure 37: Admin menu

2.2 Admin review membership form

After logging in as an admin and selecting the "Membership list" option as shown in Figure 37, you'll see a list of members and their review status (either "Reviewed" or "Not Reviewed"). To review a member's form, click the "Click Here" link next to their name. An example is given in Figure 38. For example, if you click on Hana Binti Khalid's form, you'll see her personal details like name, gender, address, job position, and salary as shown in Figure 39. Use the "Back" button to return to the list. If you click the "Next" button, it will show further details of Hana Binti Khalid. At the bottom of the form, you can choose the "Complete" button if the form is complete, or the "Incomplete" button if the form is not completed in Figure 40. Lastly, the status of the form will be sent to ALK for approval of the membership application.

REVIEW MEMBERSHIP FORM





Num	List Name	Status
1	Hana binti Khalid	Reviewed Click Here
2	Aminah binti Abu	Reviewed Click Here
3	Aisyah binti Ali	Not Review Click Here
4	Syed Mustaqim bin Syed Mustafa	Not Review Click Here
5	Abang Shinbei bin Abang Idham	Reviewed Click Here
6		
7		
8		
9		
10		
11		
12		
13		
14		

Figure 38: List of applicant's name

REVIEW MEMBERSHIP FORM







Applicant's Information

Name : <u>HANA BINTI KHALID</u>		I/C Number : <u>950122-10-0000</u>
Sex : Male Female	Religion : <u>ISLAM</u>	Marital Status : Married Single Divorced
Address : <u>NO 1 JALAN INAISARI 5, BANDAR SG LIMAU, 48000</u>		Postcode : <u>48000</u>
<u>RAWANG, SELANGOR.</u>		State : <u>Selangor</u>
Staff Number : <u>A23000001</u>	PF Number : <u>11113597</u>	Position & Grade : <u>Pengurus Kewangan, W54</u>
Office Address : <u>Ibu Pejabat KADA Lundang, 15710, Kota Bharu,</u>		Postcode : <u>15710</u> Fax/Tel Number : <u>603-5555555</u>
<u>Kelantan, Malaysia.</u>		Phone Number : <u>010-2223333</u>
House Phone Number : <u>03-60214568</u>	Monthly Salary : <u>RM 3000.00</u>	

BACK
NEXT

Figure 39: Applicant's membership form page 1

REVIEW MEMBERSHIP FORM



5

Applicant's Family Information

Num	Relationship	Name	I/C Number
1.	Father	Abdullah bin Mahmod	801203-02-1111

[Click here to add column](#)

Fees and Donations

Num	Activity	RM
1	Entrance Fee	55.00
2	Share Modal*	301.00
3	Fee Modal	100.00
4	Membership Deposit	60.00
5	Tabung Kebajikan Donation (Al-Abrar)	200.00
6	Saving	555.00
7	Others	

*Minimum share modal is RM300.00 and not exceeding 1/5 of Koperasi share modal and the payment must be made within 6 months from the date of membership approval

BACK

COMPLETE

INCOMPLETE

Figure 40: Applicant's membership from page 2

2.3 Admin review loan form

Admin start review by accessing the Review Loan Form in Figure 1. Upon logging in, admin will see a list of loan applications with their review status ("Reviewed" or "Not Reviewed") and completion status ("Complete" or "Incomplete"). The admin should click "Click Here" next to an application to begin the review process. In the Review Loan Details screen (Figure 2), verify the type of loan, loan amount, duration, and monthly repayment amount by clicking "Next" if everything is correct. Next, review the details of the first guarantor (Figure 3), confirming the guarantor's name, IC number, PF number, and latest pay slip and proceed by clicking "Next." Similarly, check the second guarantor's details (Figure 4) and ensure all information is accurate before clicking "Next." On the Applicant Agreement screen (Figure 5), verify the applicant's name and IC number and confirm the text authorizing the Koperasi to obtain necessary information and deduct repayments, then click "Next." The Guarantor Agreement screen (Figure 6) outlines the guarantor's responsibility if the borrower defaults, which should be reviewed carefully before clicking either "Complete" or "Incomplete". After completing the review of all details, return to the main review form (Figure 7) to update the review status to "Reviewed" and set the completion status to "Complete" if all steps are finalized.

REVIEW LOAN FORM

Num	List Name	Status
1	Hana binti Khalid	Reviewed Click Here
2	Aminah binti Abu	Reviewed Click Here
3	Aisyah binti Ali	Not Review Click Here
4	Syed Mustaqim bin Syed Mustafa	Not Review Click Here
5	Abang Shinbei bin Abang Idham	Reviewed Click Here
6		
7		
8		
9		
10		
11		
12		
13		
14		

Figure 41: Loan review list

REVIEW LOAN FORM

LOAN DETAILS

Type of loan : ☒ AI - Bai ☐ AI - Inah ☐ Skim Khas

☐ Baik Pulih Kenderaan ☐ Cukai Jalan ☐ Karnival Musim Istimewa

☐ Lain- lain (Nyatakan) _____

Loan Amount : RM 2200 . 00

Loan Duration : 36 month

Monthly Repayment : RM 33 . 33

NEXT

Figure 42: Application's loan form

[illegible]

NEXT

Figure 43: Guarantor 1 detail

Domo Company Sdn Bhd				NYTARAN KAWAT - 26/04/2019											
NO PERUSAHA	097100	097100		ANALISA	NO ENROLMENT										
BARANG	001001 (DOKU)			NO PAKET	120007888										
NO. WP	0000000000000000			NO KAWAT	120000000										
NAMA	OFF			NO LEMPA	120000700										
AMBER KAWAT	4.0000	30.00		NO											
				NO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PERKODANAN				POTONGAN											
GAJI ASAS		5.000.00		NO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
ALL INCLUSIVE		500.00		NO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
ATTENDANCE PROMOTIVE		50.00		NO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
ATTENDANCE INCENTIVE (30%)		200.00		NO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
DATA ATTENDANCE		100.00		NO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
INCENTIVE		200.00		NO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
MEDICAL TEST		500.00		NO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PERFORMANCE		100.00		NO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PENJUAL		100.00		NO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
ON SITE PAID		400.00		NO	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
TOTAL PERKODANAN				TOTAL PERKODANAN											
1215.00				1215.00											
DOKU BAKSI				DOKU BAKSI											
5.000.00				5.000.00											
500.00				500.00											
200.00				200.00											
100.00				100.00											
200.00				200.00											
100.00				100.00											
500.00				500.00											
100.00				100.00											
400.00				400.00											
1215.00				1215.00											
5.000.00				5.000.00											
500.00				500.00											
200.00				200.00											
100.00				100.00											
200.00				200.00											
100.00				100.00											
500.00				500.00											
100.00				100.00											
400.00				400.00											
1215.00				1215.00											
5.000.00				5.000.00											
500.00				500.00											
200.00				200.00											
100.00				100.00											
200.00				200.00											
100.00				100.00											
500.00				500.00											
100.00				100.00											
400.00				400.00											
1215.00				1215.00											
5.000.00				5.000.00											
500.00				500.00											
200.00				200.00											
100.00				100.00											
200.00				200.00											
100.00				100.00											
500.00				500.00											
100.00				100.00											
400.00				400.00											
1215.00				1215.00											
5.000.00				5.000.00											
500.00															

NEXT

Figure 44: Guarantor 2 detail

LOAN APPLICATION FORM



APPLICANT AGREEMENT

I IC NUM. :

hereby authorize KOPERASI KAKITANGAN KADA BHD or its legal representative to obtain any necessary information and also to get a refund from my salary deduction and emoluments as the amount loaned. I also agree to accept any decision from this KOPERASI to reject the application without giving any reason

BACK

NEXT

LOAN APPLICATION FORM	
GUARANTOR AGREEMENT	■ With this agreement, I/we voluntarily agree, either jointly or individually, to accept full responsibility for any loans made to cover the cost of these items or finance. If the borrower cannot return the loans in accordance with the agreed upon terms, we guarantee to reimburse all of the debt payments that were issued to him above.
COMPLETE INCOMPLETE	

ADMINISTRATOR LOGIN			
			
Num	List Name	Process	Status
1	Hana binti Khalid	Reviewed	Complete
2	Aminah binti Abu	Reviewed	Complete
3	Aisyah binti Ali	Reviewed	Incomplete
4	Syed Mustaqim bin Syed Mustafa	Not Review	Complete
5	Abang Shinbei bin Abang Idham	Reviewed	Incomplete
6			
7			
8			
9			
10			
11			
12			
13			
14			

Figure 47: Applicant’s loan list

2.4 Generate statement

Figure 48 shows the option to generate a statement for the approved loan applicant list. When the admin clicks the "Generate Statement" button, the report statement is automatically created. Figure 49 then shows that the report statement has been sent via email.

GENERATE STATEMENT



KOPERASI KAITANGAN KADA KELANTAN BERHAD
MEMBERSHIP APPLICATION FORM

Num	List Name	Status	Approved
1.	Ali bin Abu	Generate Statement Complete	☒
2.	Aminah binti Abu Bakar	Generate Statement Complete	☒
3.	Aisyah binti Ali	Generate Statement Complete	☒

Figure 48: Generate Statement

GENERATE STATEMENT



MEMBERSHIP APPLICATION FORM

Num	List Name	Status	Approved
1.	Ali bin Abu	E-mail sent! Complete	☒
2.	Aminah binti Abu Bakar	Generate Statement Complete	☒
3.	Aisyah binti Ali	Generate Statement Complete	☒

Figure 49: Report statement successfully sent via e-mail

3. ALK

3.1 ALK chose membership form

ALK is prompted to login first by entering ALK ID and Password as shown in Figure 50. Then, the ALK can choose whether to approve the membership form or loan application form in Figure 51.

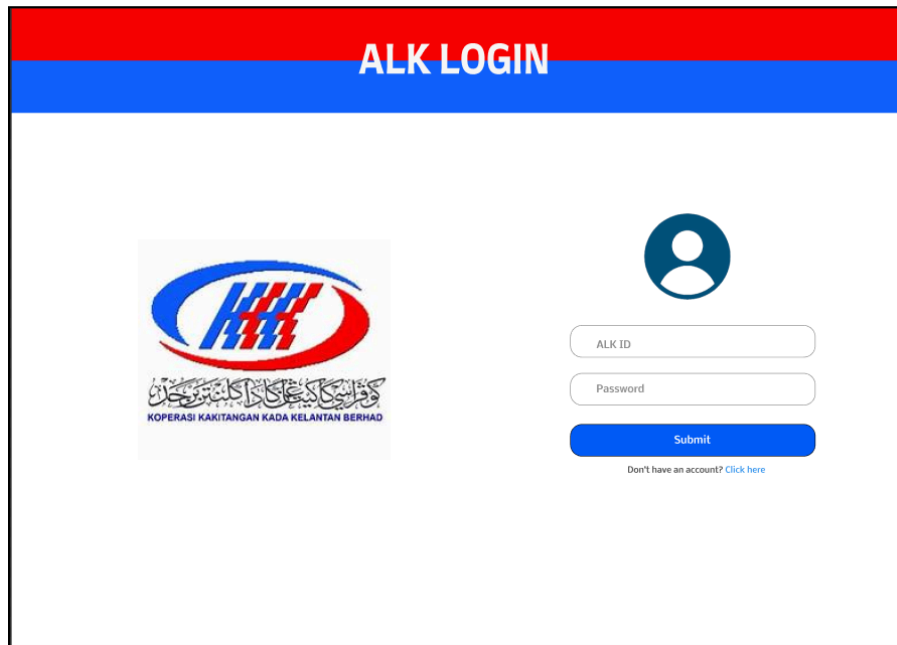


Figure 50 : ALK Login Page

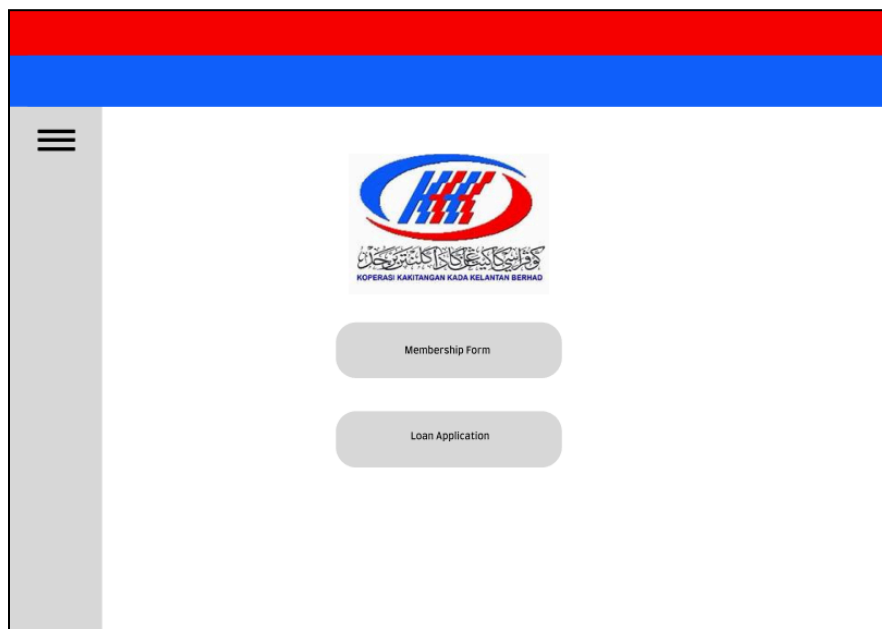
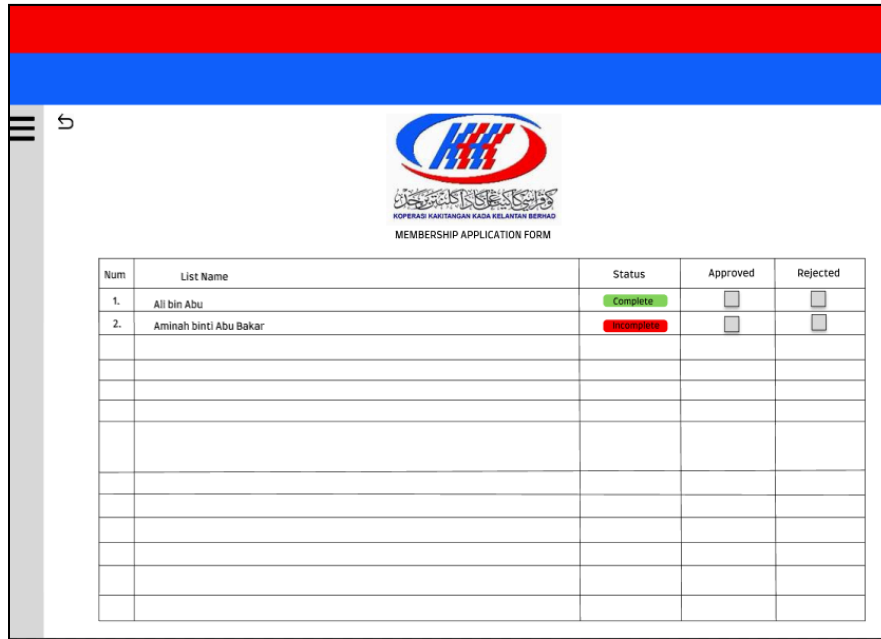


Figure 51: Choose membership Form or Loan Form

3.2 ALK choose membership form

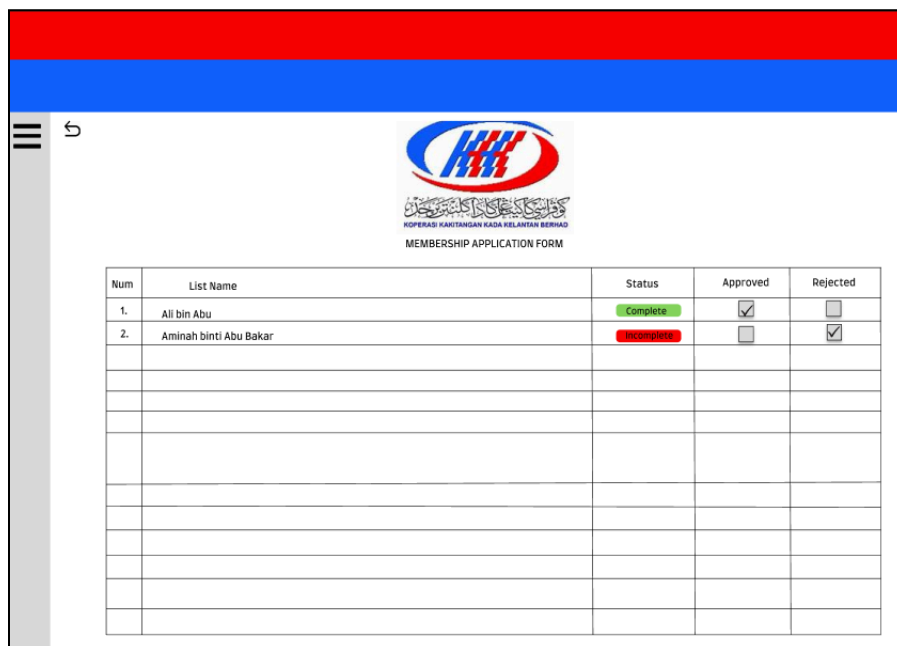
Figure 52 shows the list of names of staff that applied for membership. The ALK just needs to approve or reject by clicking the checkbox as shown in Figure 52 and the result in Figure 53.



The screenshot displays a web application interface for a membership application form. At the top, there is a red and blue header bar. Below the header, the logo of Koperasi Kuantan Kuda Kelantan Berhad is visible, along with the text "MEMBERSHIP APPLICATION FORM". The main content area contains a table with the following data:

Num	List Name	Status	Approved	Rejected
1.	Ali bin Abu	Complete	☐	☐
2.	Aminah binti Abu Bakar	Incomplete	☐	☐

Figure 52: List of Applicant Names



The screenshot displays the same web application interface as Figure 52, but with the status of the applicants updated. The table now shows the following data:

Num	List Name	Status	Approved	Rejected
1.	Ali bin Abu	Complete	☒	☐
2.	Aminah binti Abu Bakar	Incomplete	☐	☒

Figure 53: Status Page

3.3 ALK choose loan application

Figure 54 shows the list of names of KADA membership holders that applied for the loan application. Figure 55 shows the loan terms if ALK presses the “Loan Terms” button. The ALK can review each of the loan forms by clicking the “Click here” button in Figure 54. The ALK also can review the payslip by clicking “Click here” in Figure 57. The “Compare” button in Figure 58 is used to compare the applicant’s loan form with loan terms. The result is shown in Figure 60. Green indicates pass and red indicates fail. In Figure 60, ALK can make a final decision by clicking the “Approve” or “Reject” button or “Back to Homepage” button if the ALK wishes to rethink the decision. The Figure 61 shows the result if ALK directly approves or rejects the form.



Num	List Name	Status	Review	Approved	Reject
1	Hana binti Khalid	Complete	Click Here	☐	☐
2	Aminah binti Abu	Complete	Click Here	☐	☐
3	Aisyah binti Ali	Incomplete	Click Here	☐	☐
4	Syed Mustaqim bin Syed Mustafa	Incomplete	Click Here	☐	☐
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

Figure 54: List of Applicant Names

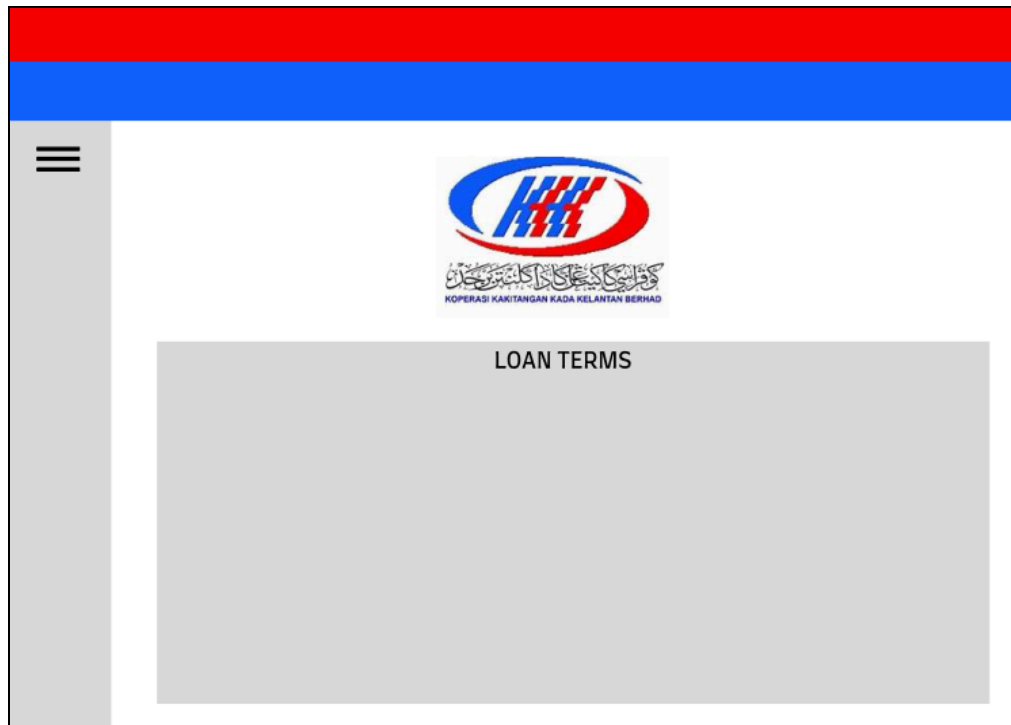


Figure 55: Loan Terms

LOAN DETAILS			
Type of loan :	AI - Bai	AI - Inah	Skim Khas
	☒ Baik Pulih Kenderaan	Cukai Jalan	Karnival Musim Istimewa
	Lain- lain (Nyatakan) _____		
Loan Amount :	RM 20,000		
Loan Duration :	24 month		
Monthly Repayment :	RM834		

[Next](#)

Figure 56: Applicant's Loan Form



GUARANTOR DETAILS

Guarantor 1 NAME : KHALID BIN AUF IC NUMBER : 000000-00-0000 PF NUM. : 88399101 LATEST PAY SLIP : Click Here	Guarantor 2 NAME : IRFAN BIN KHALID IC NUMBER : 000000-00-0000 PF NUM. : 987654 LATEST PAY SLIP : Click Here
---	---

[Next](#)

Figure 57: Applicant's Loan Form



APPLICANT LOAN DETAILS

I/C Num : 000000-00-0000		Age : 34 Years Old
D.O.B : 23 / 01 / 1990		
Sex : Female	Religion : Islam	Race : Melayu
Address : No.33, Lot 5555, Taman Mutiara, Jalan Haji Mahmud, Kelantan		
Postcode : 00000	City : Kota Bharu	
State : Kelantan		
Staff Number : 098765	PF No. : 123456	
Position : Assistant Manager		
Office Address : Bangunan KADA		
Postcode : 00000		City : Kota Bharu
Phone Number : 0134567890		
Bank Name/CWG : CIMB Bank		
Account Bank Num. : 0134567890		

[Back to Homepage](#)
[Compare](#)
[Approve](#)
[Reject](#)

Figure 58: Applicant's Loan Form



Figure 59 : Preview of Applicant's Payslip

The image shows a comparison of an applicant's loan form with loan terms. At the top, there is a red and blue header bar. Below this, on the left, is a small icon of a document with a checkmark. In the center, there is a logo for Koperasi Karyawan KADA Kelantan Berhad, featuring a stylized 'K' in blue and red. Below the logo, the word 'COMPARATION' is written in bold. The main body of the form is a table with three columns: 'Num', 'Loan Terms', and 'Status'. The table has 12 rows. The first four rows are filled with data, and the remaining eight rows are empty. At the bottom of the form, there are three buttons: 'Back to Homepage' (blue), 'Approve' (green), and 'Reject' (red).

Num	Loan Terms	Status
1	Loan term 1	☒
2	Loan term 2	☒
3	Loan term 3	☒
4	Loan term 4	☒
5		
6		
7		
8		
9		
10		
11		
12		

Back to Homepage Approve Reject

Figure 60: Comparison of Applicant's Loan Form with Loan Terms



KOPERASI KAITANGAN KADA KELANTAN BERNIAH

[Loan Term](#)

LOAN APPLICATION FORM

Num	List Name	Status	Review	Approved	Reject
1	Hana binti Khalid	Complete	Click Here	☒	☐
2	Aminah binti Abu	Complete	Click Here	☐	☐
3	Aisyah binti Ali	Incomplete	Click Here	☐	☐
4	Syed Mustaqim bin Syed Mustafa	Incomplete	Click Here	☐	☐
5					
6					
7					
8					
9					
10					
11					
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Figure 61 : Status Page